



Semantic Standardization of Unstructured Categorical Data Using Local LLMs and RAG:A Case Study on Industrial Tool Names

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Abstract

Unstructured categorical data generated by manual input leads to poor data quality and reduced machine learning performance due to lexical diversity. This study proposes a semantic standardization pipeline that combines open-source giant language model (LLM) and search augmentation generation (RAG). Twelve open-source LLMs were evaluated in two RAG configurations on Seoul's tool rental. BM25+FAISS hybrid search, candidate enlargement and RAG with category injection achieved the accuracy of up to 88.47%. To check the contribution of RAG to LLM-only +61pp or higher: It was quantitatively found that LLM alone was only 9~27% without RAG, which is below the practical level. As a result, it was experimentally confirmed that domain-specific search architecture has a greater impact on performance than model size.

Introduction

Background

- In data-intensive industries, manually entered categorical fields frequently yield “**Dirty data**”: inconsistent, non-standard representations of the same concept.

Item	Value
Seoul Metropolitan Tool Rental Open Dataset (public)	11,525
Dirtydata of tool name	3,352
Standard tool name categories	160

Problem Statement

- Manual standardization of 3,352 records requires 3~5 expert working days.
- Rule-based method demands hundreds of hand-crafted rules, with continuous maintenance as new entries arrive.
- Limitation of Rule-based method
 - Regular expression, morpheme analyzer → Syntactic errors only
 - Semantic context incomprehensible
 - Remaining Japanese: 'Giri (ぎり)' → 'Drill bit' cannot be converted
 - specification mixed: 'BOSCH 18V charging drill set' → 'drill'? 'drill set'?
 - Need hundreds of rules for 3,352 variants → High maintenance costs

Contributions

- 1.**RAG pipeline** Korean-specialized embedding, hybrid retrieval, candidate pool expansion, category injection
- 2.**Comparative benchmark** — Experimental performance comparison of 9 open-source LLM models.
- 3.**Dirty data type analysis** — per-type accuracy across 11 dirty data categories, identifying failure mechanisms for each type.
- 4.**LLM-only vs. RAG quantitative analysis** — zero-shot LLM-only experiment isolating and measuring the precise contribution of RAG retrieval.

Materials and Methods

Dirtydata Type	Dirtydata Name	Values
Attribute-embedded	"180mm Sander", "Drill (battery)"	946
Brand inclusion	"BOSCH push drive"	209
Power source specified	"Cordless electric drill (20V)"	330
Japanese-derived terms	"giri-set", "ppanchi"	87
Typos / encoding errors	"gyeongrang monkey"	117

Table. 1. Example of Dirtydata Types

Retrieval DB

- 160 standard tool names

Evaluation Dataset

- 457 curated(Dirtydata Name => Standard name) pairs

Component	Specification
Embedding model	BGE-m3-ko (1024-dim, Korean)
Retrieval	BM25 (40%) + FAISS (60%) hybrid
Candidate pool (k)	15
Prompt	Category injection (tool category & subcategory)
LLM Tool	Ollama
OS	Windows 11 + WSL2(Ubuntu)
GPU	NVIDIA RTX 5080(VRAM 17.1GB)

Table. 2 System Environments

Family Category	Models	Parameters (B)
Gemma	gemma3:4b, gemma4:e2b, gemma4:e4b	4 / 2 / 4
Granite	granite4.1:3b, granite4.1:8b	3 / 8
EXAONE	exaone3.5:2.4b, exaone3.5:7.8b	2.4 / 7.8
DeepSeek	deepseek-r1:1.5b, deepseek-r1:8b	1.5 / 8

Table. 3 Model Evaluated

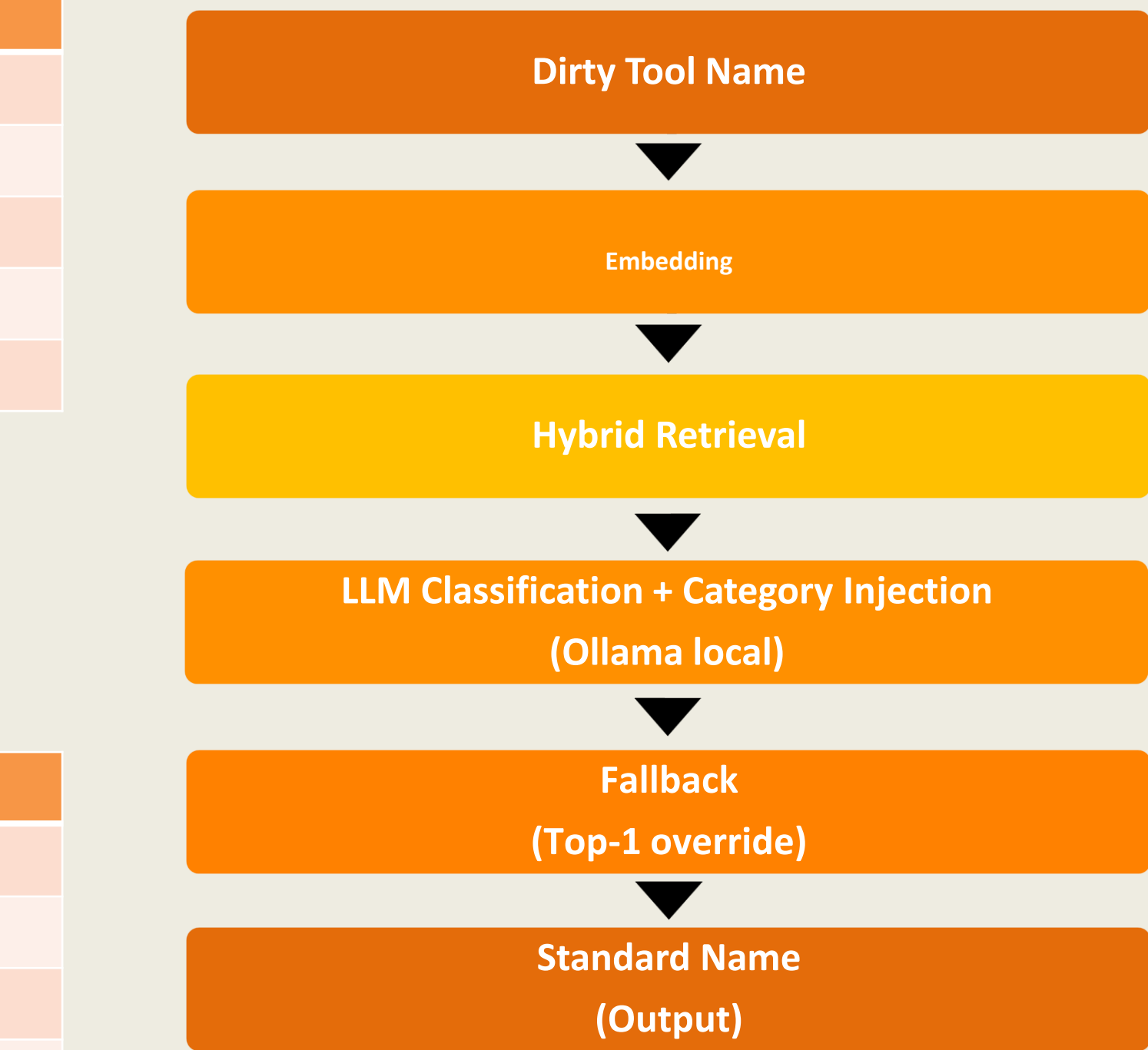
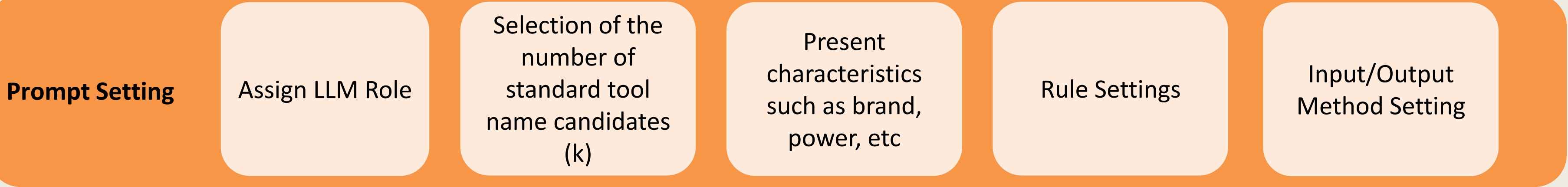


Fig. 3. RAG Pipeline Architecture

Metric	Definition
Accuracy	Exact match rate (Predicted Standard name vs Ground Truth)
Fallback Rate	Percentage of LLM outputs outside candidate list
Processing Speed	Time for LLM to retine 1 name(Item/second)
Total LLM time	Time for LLM to refine 3,352 names

Table. 4 Evaluation Metrics



Results

1 Performance of Local LLM Models

Rank	Model	Fallback (%)	Speed (Item/s)	LLM Time (min)
1	Gemma4:e4b	8.1	1.58	35
2	Gemma3:4b	6.5	1.77	32
3	Granite4.1:8b	5.0	1.21	46
4	Gemma4:e2b	6.4	1.93	29
5	Exaone3.5:7.8b	9.2	1.23	46
6	Deepseek-r1:1.5b	85.2	1.49	37
7	Deepseek-r1:8b	2.6	1.40	40
8	Exaone3.5:2.4b	13.6	1.58	35
9	Granite4.1:3b	20.8	1.53	36

table. 5 Overall Performance

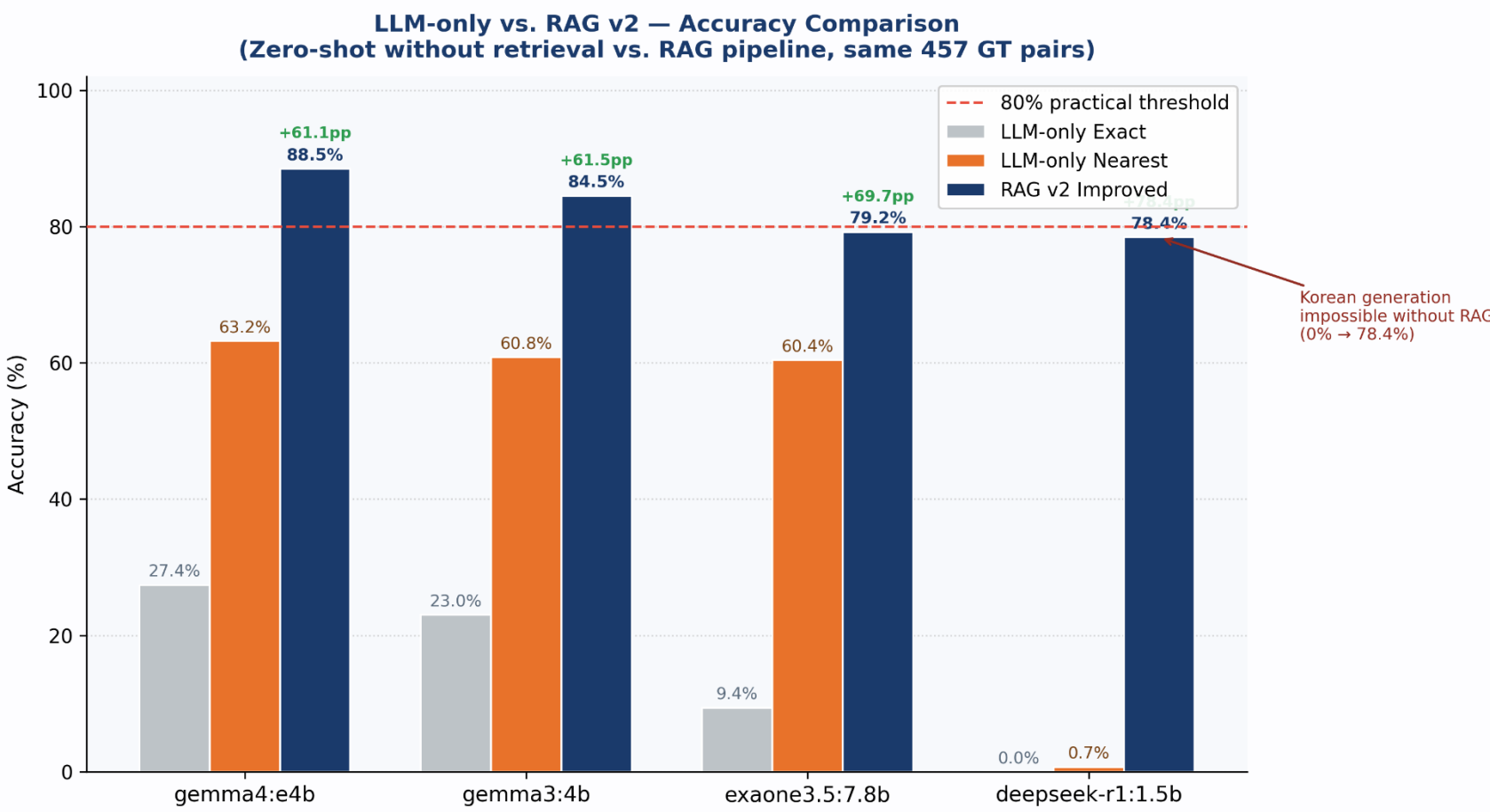


Fig. 3 LLM-only VS RAG – Accuracy Comparison
Exact: output matches ground-truth label exactly. **Nearest:** output is semantically close but does not match the precise standard notation.

2 Key Findings

LLM Model Comparison	Results
Gemma3:4b (3.3GB) vs Exaone3.5:7.8b (4.8GB)	84.5% > 79.2%
Deepseek-r1:1.5b (1.1GB) vs Deepseek-r1:8b (5.2GB)	78.1% ≈ 77.8%

table. 6 Comparison of performance according to the parameters of the model

Fig. 1 Overall Accuracy Performance Graph

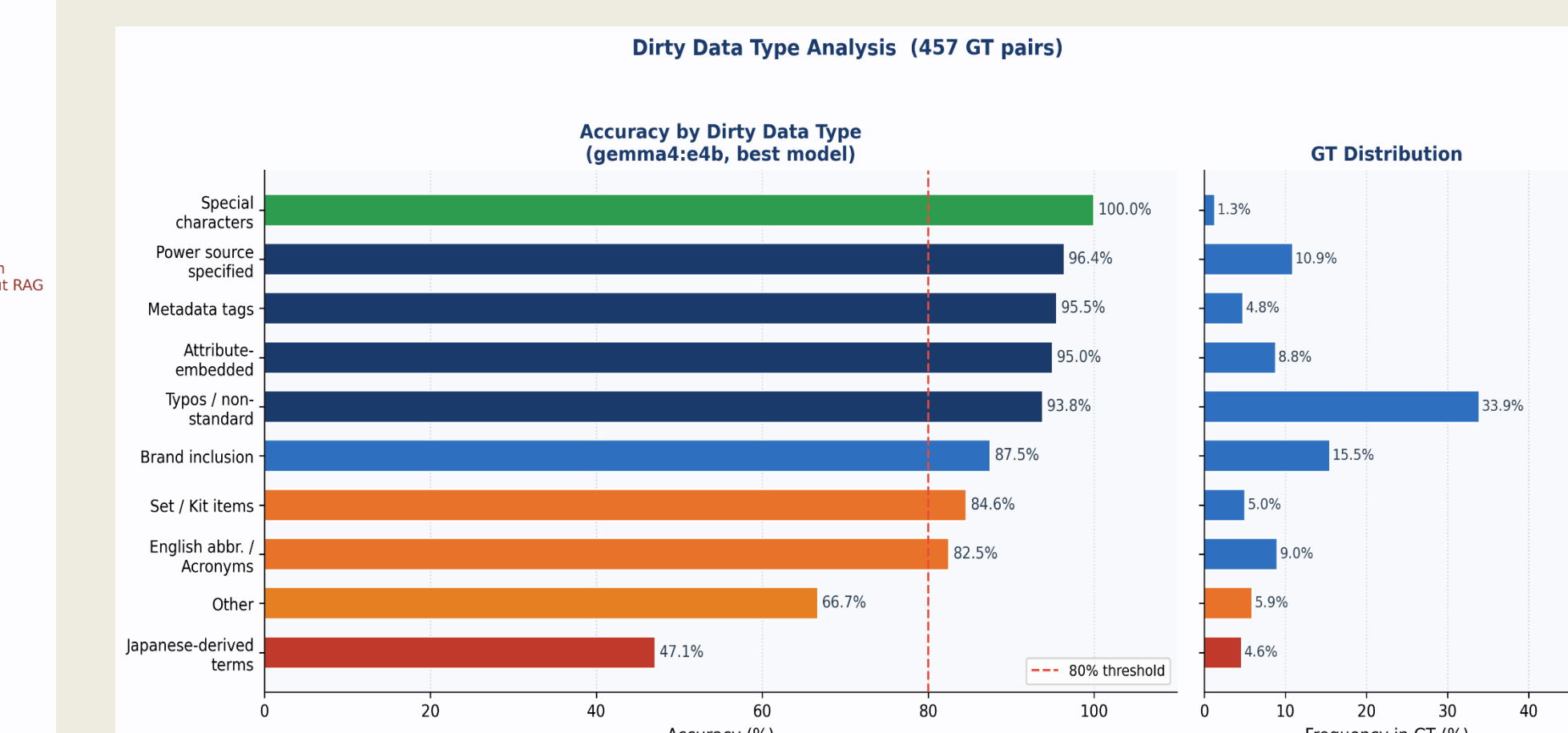


Fig. 2 Dirty Data Type Analysis for highest performance LLM model(Gemma4:e4b) and Ground Truth(GT) Distribution

Process	Performance
Embedding Time	About 10sec(3,352 batch)
LLM processing Time	1.2~2.0/sec
Total processing Time	30~50min

table. 7 Performance by Process

Model size does not predict performance in RAG settings

- EXAONE (specialized in Korean) has low performance in this task(< 80%).
- The Gemma family shows excellent performance overall(> 80%).
- RAG-supplied domain knowledge compensates for limited model capacity
- Conformity with constrained output formats over domain knowledge + ability to infer candidate comparisons determine performance.

Dirty Data Types Explain Performance Gap

- Typos is the most common type of total GT and achieves 93.8% accuracy.
- The largest type of contribution among missing RAG candidates is Japanese derived terms. All nine models are consistently low, ranging from 23.5 to 52.9%

Process Time

- In the RTX 5080 environment, the processing time is as fast as 30 to 50 minutes.

Conclusions

1 RAG-Based Pipeline for Semantic Standardization

A RAG-based semantic standardization pipeline for unstructured categorical dirty data, validated in real Korean industrial tool name records.

2 Four Targeted Improvements Yield 88.47% Accuracy

Korean-specialized embedding, BM25+FAISS hybrid retrieval, expanded candidate pool(k=15), and category-augmented prompting collectively raise accuracy.

3 RAG Is Indispensable: +61 to +78 pp Over LLM-only

A zero-shot LLM-only experiment confirms that RAG is not merely helpful but indispensable. LLMs alone achieve only 0~27% exact-match accuracy. RAG serves as an output anchor and enables models with limited language generation capability to achieve practical performance through candidate selection.

4 Compact Models Match Larger Ones with Domain-Optimized Retrieval

Smaller models with domain-optimized retrieval achieve competitive or superior accuracy compared to larger models, demonstrating the cost-effectiveness of the proposed approach.

